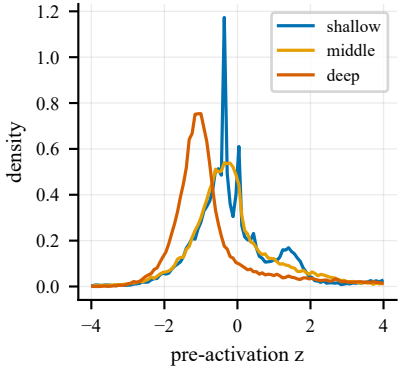
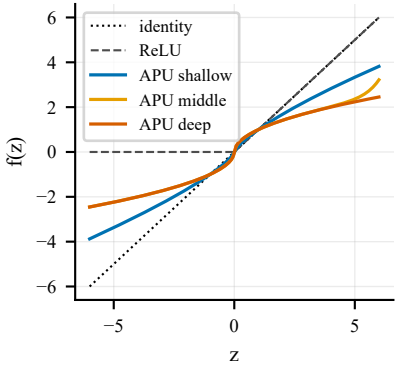
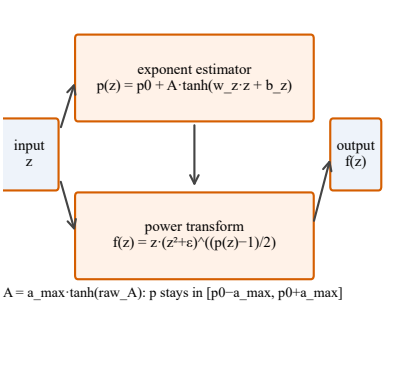

(a) feature statistics differ by depth
shallow
middle
deep
density
pre-activation z
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−4
−2
0
2
4


(b) learned APU shapes
identity
ReLU
APU shallow
APU middle
APU deep
f(z)
z
6
4
2
0
−2
−4
−6
−5
0
5


(c) APU computation
exponent estimator
p(z) = p0 + A·tanh(w_z·z + b_z)
input
z
output
f(z)
power transform
f(z) = z·(z²+ε)^((p(z)−1)/2)
A = a_max·tanh(raw_A): p stays in [p0−a_max, p0+a_max]